

LAB EXPERIMENT RECORD

SET 6 — Product Sales Analysis

Department	AIDS
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R
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A. Aim

To analyze monthly product sales using **grouped bar charts, stacked area charts, tables, and an interactive dashboard** in R/Tableau.

B. Dataset

Product ID	Product Name	January Sales	February Sales	March Sales
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

Program 1 — Grouped Bar Chart

Code

```
product <- c("Product A", "Product B", "Product C")
```

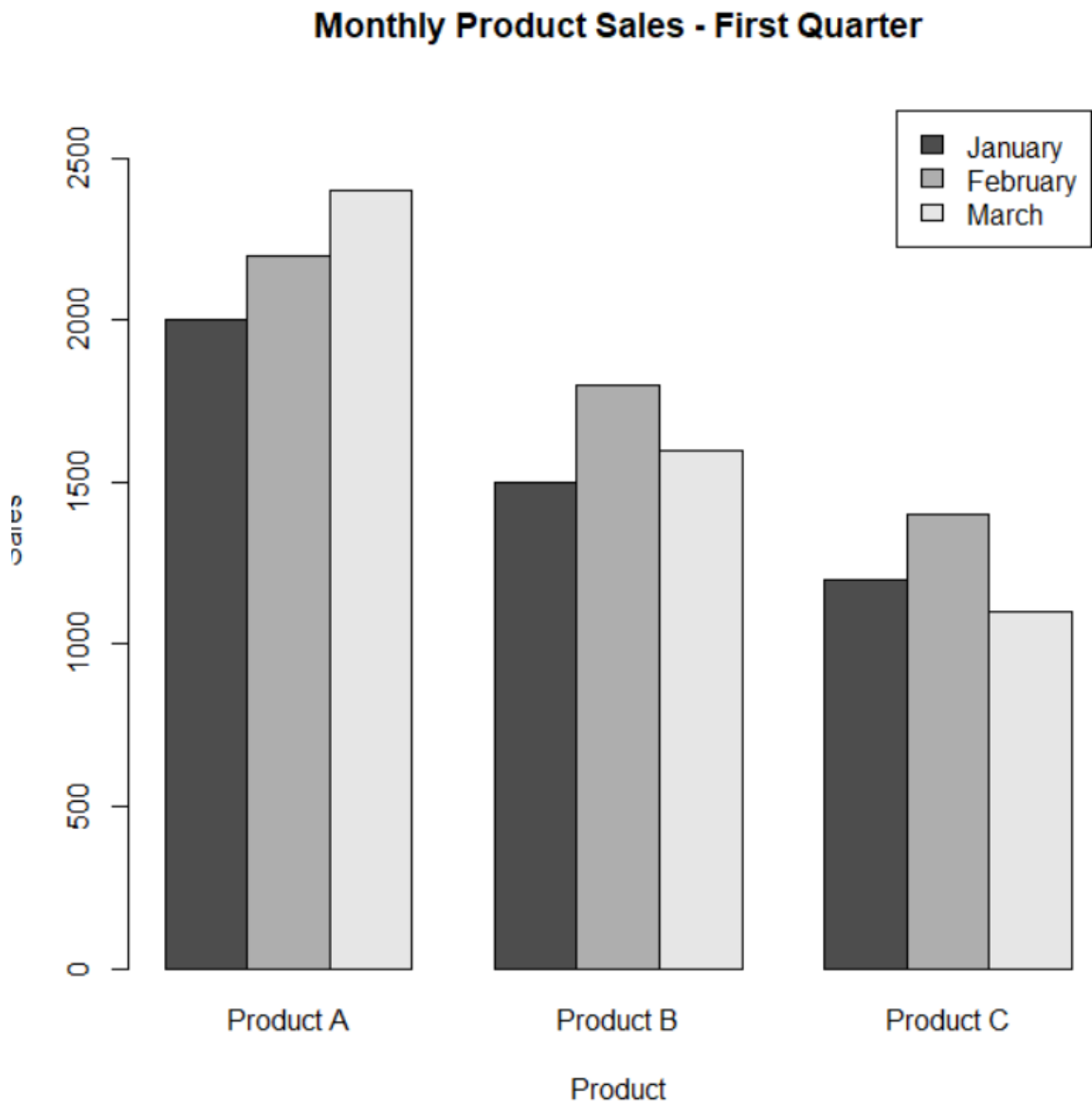
```
january <- c(2000, 1500, 1200)
```

```
february <- c(2200, 1800, 1400)
```

```
march <- c(2400, 1600, 1100)
```

```
sales <- rbind(  
  January = january,  
  February = february,  
  March = march  
)
```

```
barplot(  
  sales,  
  beside = TRUE,  
  names.arg = product,  
  main = "Monthly Product Sales - First Quarter",  
  xlab = "Product",  
  ylab = "Sales",  
  legend.text = TRUE,  
  ylim = c(0, 2700)  
)
```



Observation

Product A has the highest sales in all three months. Its sales increase from **2000 in January** to **2400 in March**. Product C has the lowest March sales at **1100**.

Program 2 — Stacked Area Chart

This is a **true stacked area chart**, showing the contribution of each product to total sales over the three months.

Code

```
month <- c("January", "February", "March")

product_A <- c(2000, 2200, 2400)
product_B <- c(1500, 1800, 1600)
product_C <- c(1200, 1400, 1100)

sales <- cbind(product_A, product_B, product_C)

total1 <- sales[, 1]
total2 <- sales[, 1] + sales[, 2]
total3 <- sales[, 1] + sales[, 2] + sales[, 3]

x <- 1:3

plot(
  x,
  total3,
  type = "n",
  xaxt = "n",
  main = "Stacked Area Chart of Product Sales",
  xlab = "Month",
  ylab = "Sales",
  ylim = c(0, max(total3))
)
```

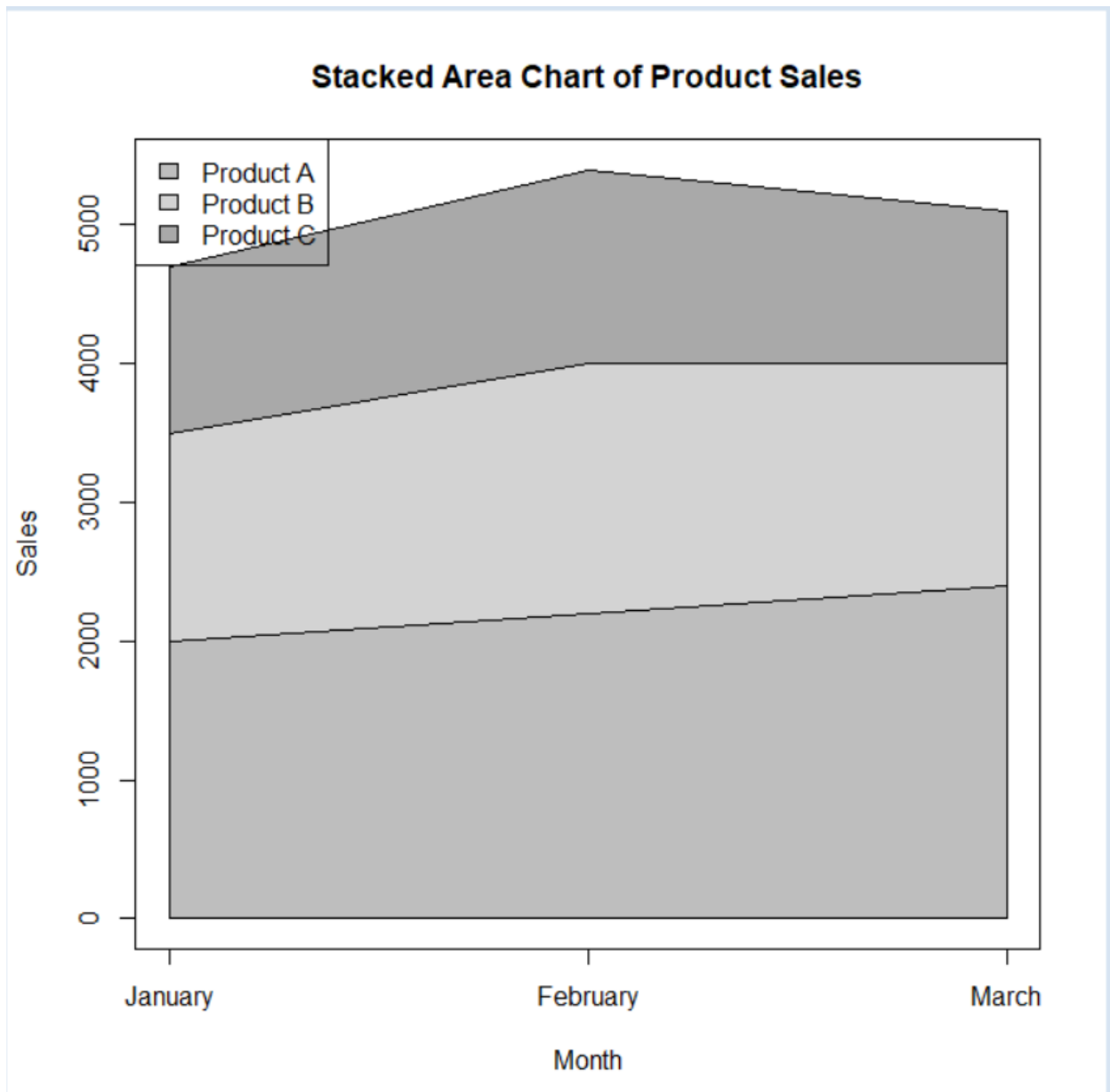
```
axis(1, at = x, labels = month)
```

```
polygon(  
  c(x, rev(x)),  
  c(total2, rev(total1)),  
  col = "lightgray"  
)
```

```
polygon(  
  c(x, rev(x)),  
  c(total1, rep(0, 3)),  
  col = "gray"  
)
```

```
polygon(  
  c(x, rev(x)),  
  c(total3, rev(total2)),  
  col = "darkgray"  
)
```

```
legend(  
  "topleft",  
  legend = c("Product A", "Product B", "Product C"),  
  fill = c("gray", "lightgray", "darkgray")  
)
```



Observation

The chart shows the contribution of Products A, B, and C to overall sales. **Product A contributes the largest portion**, while Product C contributes the smallest portion in March.

Program 3 — Monthly Sales Table

Code

```
sales_data <- data.frame(  
  Product = c("Product A", "Product B", "Product C"),  
  January = c(2000, 1500, 1200),  
  February = c(2200, 1800, 1400),  
  March = c(2400, 1600, 1100)  
)
```

```
print(sales_data)
```

```
└─ /  
>  
> print(sales_data)  
  Product January February March  
1 Product A    2000     2200  2400  
2 Product B    1500     1800  1600  
3 Product C    1200     1400  1100  
>  
>  
> |
```

Observation

The table displays the monthly sales values of all three products. Product A records the highest sales throughout the first quarter.

Program 4 — Tableau Dashboard

Step 1 — Create Grouped Bar Chart

1. Open Tableau.
2. Connect the sales dataset.
3. Create a new worksheet.
4. Drag **Product Name** → Columns.
5. Drag **January Sales, February Sales, March Sales** → Rows.
6. Select **Bar Chart**.
7. Set **Analysis** → **Stack Marks** → **Off**.
8. Add labels using **Show Mark Labels**.
9. Rename the sheet **Quarterly Sales Bar Chart**.

Step 2 — Create Stacked Area Chart

1. Create another worksheet.
2. Place **Month** on Columns.
3. Place **Sales** on Rows.
4. Select **Area Chart**.
5. Place **Product Name** on Color.
6. Keep stacking enabled.
7. Rename the sheet **Product Sales Trend**.

Step 3 — Create Sales Table

1. Create another worksheet.
2. Drag **Product Name** to Rows.
3. Drag **January Sales, February Sales, March Sales** to the view.
4. Rename it **Monthly Sales Table**.

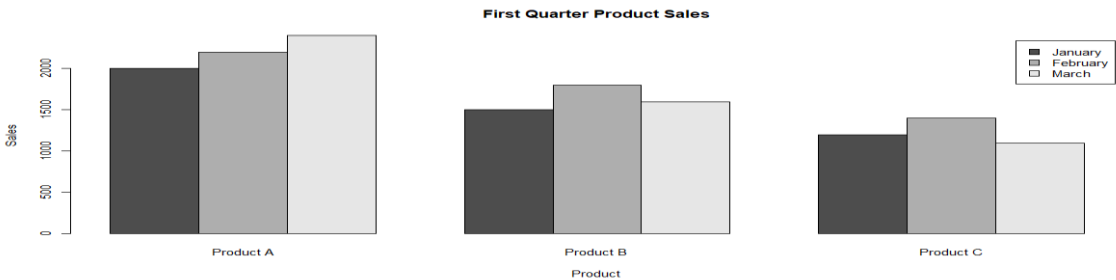
Step 4 — Create Dashboard

- 1. Select **New Dashboard**.
- 2. Drag the **Quarterly Sales Bar Chart** into the dashboard.
- 3. Drag the **Product Sales Trend** below/next to it.
- 4. Drag the **Monthly Sales Table** into the dashboard.
- 5. Add filters if required.
- 6. Give the dashboard the title:

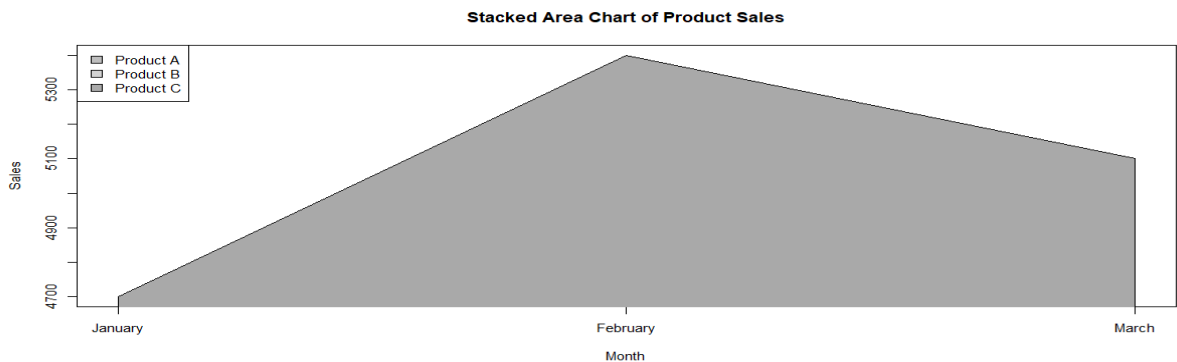
Product Sales Analysis Dashboard

Product Sales Analysis Dashboard

Grouped Bar Chart



Stacked Area Chart



Monthly Sales Table

Product	January	February	March
Product A	2000.00	2200.00	2400.00
Product B	1500.00	1800.00	1600.00
Product C	1200.00	1400.00	1100.00

C. Overall Procedure

1. Enter the product sales dataset in R/Tableau.
2. Create a grouped bar chart to compare quarterly sales.
3. Create a stacked area chart to visualize sales contribution and trend.
4. Display monthly sales using a table.
5. Combine the visualizations into an interactive Tableau dashboard.
6. Analyze and compare the sales performance of each product.

D. Result

The monthly sales of Products A, B, and C were successfully analyzed using **grouped bar charts, stacked area charts, tables, and a Tableau dashboard**. Product A showed the highest overall sales during the first quarter.

E. Key R Functions Used

Function	Purpose
c()	Create vectors
rbind()	Combine rows
cbind()	Combine columns
barplot()	Create grouped bar chart
plot()	Create basic plot
polygon()	Create stacked area regions
axis()	Add/customize axes
legend()	Add chart legend
data.frame()	Create table/data frame
print()	Display data